Smart-Camera-DF511-SL — Control4 Driver

Overview

Complete Control4 driver for **Slomins DF511-SL Camera** with full API integration, authentication, streaming, lock control, and real-time MQTT event detection.

Version: 2
Package: Slomins-doorvideolock-DF511.c4z (22.1 KB)
Minimum Control4 OS: 3.3.2+

Features

✅ Full API Integration

- Initialize camera and retrieve public key
- User authentication with RSA-OAEP encryption
- Token management and device listing

✅ Streaming Support

- RTSP main stream (high quality)
- RTSP sub stream (low quality)
- Dynamic URL generation

✅ Camera Functions

- Snapshot URL generation
- Preset management

✅ Lock Control

- Remote lock and unlock commands via Cloudbus API
- Full Control4 Lock Proxy integration
- Supports Control4 proxy commands:
  - LOCK
  - UNLOCK
  - QUERY_STATE
- Automatic Control4 UI padlock updates
- Lock status synchronization with Control4 automation
- Lock/Unlock events fired for Composer automation

✅ Auto Relock

- Optional automatic relock after unlocking
- Configurable **Lock Seconds** property
- Existing timers are cancelled when a new unlock occurs
- Automatically triggers lock command after timer expires

✅ Security

- RSA-OAEP + SHA256 encryption using `C4:Crypto()`
- HMAC-SHA256 signatures for API requests
- Secure token storage

✅ Real-Time Event Detection (MQTT)

- Motion detection events
- Human detection events
- Stranger detection alerts
- Camera online/offline monitoring
- Snapshot attachment support for notifications

Requirements

- **Control4 Composer Pro**
- **Control4 OS 3.3.2+**
- **Network access** to API endpoint and camera
- Camera accessible on LAN

Installation

1. Install Driver Package

1. Open **Control4 Composer Pro**
2. Go to **Drivers** → **Add Driver** → **Install From File**
3. Select `Slomins-doorvideolock-DF511.c4z`
4. Click **Install**
5. Driver will appear in the available drivers list

2. Add Device to Project

1. Open **Project View**
2. Select a room
3. Click **Add Device**
4. Search for **Slomins DF511 Door Video Lock**
5. Add to the room

3. Configure Properties

API Configuration

- **Base API URL:** `https://api.arpha-tech.com`
- **Account:** Your email or phone number

Camera Configuration

- **IP Address:** Camera IP (example `192.168.1.100`)
- **HTTP Port:** 80
- **RTSP Port:** 554
- **Username:** Camera username
- **Password:** Camera password

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4. Enable Lock Tile in Control4 UI

To make the **DF511 Lock tile visible** in the Control4 interface:

1. Open **Composer Pro**
2. Navigate to the room where the device was added
3. Select the **Security** tab
4. Locate **DF511 Lock**
5. Enable **Show in Navigator**
6. Click **OK**

The lock control tile will now appear in the Control4 mobile app and touchscreens.

Quick Start Guide

Complete Setup Flow

```
1. Execute "Initialize"
   → Gets RSA public key from API
   → Stores in "Public Key" property

2. Execute "Login/Register"
   → Encrypts credentials with RSA-OAEP
   → Authenticates with API
   → Stores Auth Token

3. Execute "Test Main Stream"
   → Generates RTSP URL for streaming

4. Execute "Get Snapshot URL"
   → Generates HTTP snapshot URL

5. Test Lock Control
   → Lock/Unlock using Control4 Navigator
   → Driver sends commands to API
   → Lock status updates automatically
```

🌐 Dynamic IP Auto-Update (Online Event Handling)

The driver includes intelligent handling of camera connectivity to ensure the correct IP address is always used.

🔄 How It Works

Whenever the camera comes online, the driver automatically:

- Detects the **Camera Online** event
- Calls the **Get Devices API**
- Matches the device using **VID (Virtual ID)**
- Retrieves the latest **local IP address**
- Updates the **IP Address** property in the driver
- Pushes the updated address to the **Control4 Camera Proxy**

🔔 Real-Time Event Detection

The driver supports **MQTT-based real-time event streaming over SSL (port 8884)** and provides notifications for the following events:

- Doorbell ring notifications with snapshot
- Motion detection with snapshot attachment
- Human detection
- Stranger detection
- Camera online/offline status monitoring

MQTT Configuration

Default Behavior

MQTT is **enabled automatically by the driver after authentication**.

The driver will automatically:

1. Enable MQTT
2. Fetch MQTT credentials
3. Connect to the MQTT broker
4. Subscribe to camera event topics

You normally **do not need to enable MQTT manually**.

CldBus App Motion Detection Settings

To receive **Motion Detection** and **Human Detection** events, detection must be enabled in the **CldBus mobile app**.

Steps

1. Open the **CldBus App**
2. Select your **camera device**

3. Tap **Settings**
4. Navigate to:

```
Settings → Motion Detection
```

5. Under **Detection Type**, select one of the following:

Detection Type	Description
All Detections	All movement events will trigger notifications. This includes general motion detection.
Human Detection	Only human movement will trigger events. Non-human motion will be ignored.

Push Notification Configuration

1. Create Push Notification

1. Open **Composer Pro**
2. Navigate to:

```
Agents → Push Notification
```

3. Click **Add Notification** then enter a name for the notification.

Configure the following:

Setting	Value
Category	Cameras
Severity	Info / Critical
Subject	Camera Event

Click **Save**.

2. Enable Snapshot Attachment

Edit the created notification and set:

```
Attachment Type = Snapshot URL
```

This allows push notifications to include the **camera snapshot image**.

Mapping Push Notifications in Programming

1. In **Composer Pro**, open **Programming**. 2. Select the **Camera Driver** from the device list. 3. In the **Events** section, choose the event you want to trigger the notification (for example **Motion Detected**).

On the **right-side panel**:

4. Click **Push Notifications**. 5. From the dropdown list, select the **notification you created earlier**. 6. Drag and drop the notification into the programming area **or double-click the green arrow** to add it.

This maps the camera event to the push notification.

Example:

WHEN Motion Detected THEN Send Push Notification

Notification Flow

```
Camera Event (MQTT)
        ↓
Driver Receives Event
        ↓
Driver Processes Event
        ↓
Control4 Event Triggered
        ↓
```

```
Push Notification Agent
        ↓
Mobile Notification Sent
        ↓
Snapshot Image Attached
```

Available Actions

Accessible via **Device → Actions**

Authentication

- Initialize
- Login/Register
- Get Devices

Streaming

- Test Main Stream
- Test Sub Stream
- Get Snapshot URL

Lock Controls

- Lock
- Unlock

Properties

Input Properties

Property	Type	Default	Description
Base API URL	STRING	https://api.arpha-tech.com	API endpoint
Account	STRING	-	User email/phone
IP Address	STRING	192.168.1.100	Camera IP
HTTP Port	INTEGER	3333	HTTP port
RTSP Port	INTEGER	554	RTSP port
Username	STRING	-	Camera username
Password	STRING	-	Camera password
Snapshot Path	STRING	/wps-cgi/image.cgi	Snapshot path
Enable MQTT	LIST	True	Enable/disable MQTT event streaming

Output Properties

Property	Description
Status	Current driver status
Public Key	RSA public key
Client ID	Generated client ID
Auth Token	Authentication token
Main Stream URL	High quality RTSP URL
Sub Stream URL	Low quality RTSP URL
Lock State	Locked / Unlocked
Online	Camera online status

API Functions

1. Initialize Camera

Command: `INITIALIZE_CAMERA`

Endpoint: `POST /api/v3/openapi/init`

Purpose: Retrieve RSA public key for encryption

What it does:

- Generates client ID and request ID
- Creates HMAC-SHA256 signature
- Requests public key from API
- Stores key in "Public Key" property

Expected Output:

```
Camera initialization succeeded
Received public key: MIGfMA...
Public key stored successfully
```

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2. Login/Register

Command: `LOGIN_OR_REGISTER`

Endpoint: `POST /api/v3/openapi/auth/login-or-register`

Purpose: Authenticate user with encrypted credentials

Prerequisites:

- Must run Initialize first (needs public key)
- Account must be set in properties

What it does:

- Encrypts credentials with RSA-OAEP + SHA256
- Sends authentication request
- Stores auth token and user ID
- Token used for subsequent API calls

Encryption Process:

```
1. Create crypto object: C4:Crypto("RSA")
2. Import public key
3. Encrypt JSON: {"country_code":"US","account":"user@email.com"}
4. Generate HMAC-SHA256 signature
5. Send to API
```

Expected Output:

```
Using C4:Crypto for RSA-OAEP encryption...
Crypto object created successfully
Public key imported successfully
Encrypting data with RSA-OAEP + SHA256...
Encryption successful!
Login/Register succeeded
Auth token stored
User ID: 12345
```

3. Test Main Stream

Command: `TEST_MAIN_STREAM`

Purpose: Generate high quality RTSP stream URL

Format: `rtsp://<ip>:554/streamtype=1`

Prerequisites:

- IP Address set in properties
- RTSP Port (default: 554)

Example Output:

```
Main Stream RTSP URL: rtsp://192.168.1.100:554/streamtype=1
```

Result:

- URL stored in "Main Stream URL" property
- Use URL in VMS or video player

---

4. Test Sub Stream

Command: `TEST_SUB_STREAM`

Purpose: Generate low quality RTSP stream URL

Format: `rtsp://<ip>:554/streamtype=0`

Prerequisites:

- IP Address set in properties
- RTSP Port (default: 554)

Example Output:

```
Sub Stream RTSP URL: rtsp://192.168.1.100:554/streamtype=0
```

Result:

- URL stored in "Sub Stream URL" property
- Use URL in VMS or video player

---

5. Get Snapshot URL

Command: `GET_SNAPSHOT_URL`

Purpose: Generate HTTP snapshot URL

Format: `http://<ip>:<port>/wps-cgi/image.cgi?resolution=640x480`

Prerequisites:

- IP Address set in properties

- Optional: Username/Password for authentication

Example Output:

```
Snapshot URL: http://192.168.1.100:3333/wps-cgi/image.cgi?resolution=640x480
```

6. Lock Status & Control

Purpose: Control and monitor the door lock integrated in the DF511 device.

API Endpoint Example

```
GET /v1.0/devices/{deviceId}/status
```

What it does:

- Retrieves the device status from the cloud API
- Extracts the `lock_motor_state` value
- Updates Control4 lock proxy
- Synchronizes lock state with Control4 UI

Driver Behavior

When the driver receives lock status from the API:

API Value	Control4 State
true	Locked
false	Unlocked

The driver then updates the Control4 lock proxy:

```
C4:SendToProxy(5002, "LOCK_STATUS_CHANGED", { LOCK_STATUS = "locked" })
```

or

```
C4:SendToProxy(5002, "LOCK_STATUS_CHANGED", { LOCK_STATUS = "unlocked" })
```

This ensures the **Navigator lock icon updates correctly**.

Automation Events

The driver fires Control4 events when state changes:

- `Lock`
- `Unlock`
- `Lock Error`
- `Unlock Error`

Logging & Troubleshooting

View Logs

1. Open **Composer Pro**
2. Go to **Lua Output**
3. Execute driver actions
4. Review request and response logs

Common Issues

No public key available

- Run Initialize first
- Check API URL

Login failed

- Verify Account property
- Ensure Initialize ran successfully

IP Address not set

- Verify camera is reachable on LAN

No auth token available

- Run Login/Register first

MQTT not receiving events

- Verify **Enable MQTT = True**
- Confirm authentication completed
- Ensure motion detection enabled in CldBus app
- Verify outbound port **8884** is allowed

API Response Codes

- **200 / 20000** - Success
- **401** - Unauthorized (invalid token)
- **400** - Bad request (invalid parameters)
- **500** - Server error

Building the Driver

File Structure

```
Slomins-doorvideolock-DF511/
├── driver.lua
├── driver.xml
├── mqtt_manager.lua
├── README.md
├── CldBusApi/
│   ├── auth.lua
│   ├── dkjson.lua
│   ├── http.lua
│   ├── sha256.lua
│   ├── transport_c4.lua
```

```
│   └── util.lua
└── build-c4z.ps1
```

Build Package

Run:

```
.\build-c4z.ps1
```

This generates:

```
Slomins-doorvideolock-DF511.c4z
```

Development Notes

Helper Libraries (CldBusApi/)

- **auth.lua** - Authentication utilities
- **dkjson.lua** - JSON encoding/decoding
- **http.lua** - HTTP request helpers
- **sha256.lua** - HMAC-SHA256 implementation
- **transport_c4.lua** - Control4 transport adapter
- **util.lua** - UUID generation, HMAC functions

Signature Generation

All API requests include HMAC-SHA256 signatures:

```
-- Message format
local message = "client_id=<uuid>&request_id=<uuid>&time=<timestamp>&version=0.0.1"
```

```
-- Generate signature
local signature = util.hmac_sha256_hex(message, app_secret)

-- App secret
local app_secret = "hg4IwDpf2tvbVdBGc6nwP5x2XGCIlNv8"
```

UUID Generation

```
local client_id = util.uuid_v4()
-- Example: "0ffb133a-b03d-4cfd-82a9-d45220516199"
```

Version History

v3.0 (Current)

- Implemented functionality to store all incoming events in the database.

v2.0

- Fixed version display showing 0 in Control4 Manage Drivers interface
- Fixed device specific conditionals labels not showing in Control4 Composer

v1.0.0

- Complete API integration
- RSA-OAEP encryption
- Authentication and token management
- Device listing
- RTSP streaming
- Snapshot support
- Lock proxy integration
- MQTT real-time event detection
- Push notification support

v1.0.1

- HTML Readme File
- Dynamic credentials retrieval

v1.0.2

- Manually Locking and locking trigger events/Scenes in Control4

v1.0.3

- Implemented real-time state synchronization for lock/unlock status to ensure the UI reflects changes instantly without requiring a refresh.

v1.0.4

- Bug Fixes : device hang when API is unreachable, MQTT connection stability improvements

v1.0.5

- Dynamic Cldbus API Id and App Secret retrieval from API when MAC address is provided

v1.0.6

- Added conditionals in composer pro for mapping lock and unlock events to push notifications, ensuring that notifications are only sent when the corresponding event occurs.
- Disabled ptz on the command panel of the composer pro as the camera does not support ptz, preventing confusion and improving usability for users configuring the driver in their Control4 system.

v1.0.7

- Improved device wake-up reliability by implementing a retry mechanism for API calls when the camera is offline.
- Bug Fixes: Reinitializes the device when account property changes to ensure the driver is fetching the latest authentication token.

v1.0.8

- Bug Fixes: Low Battery, Battery Warning, and Battery Critical events are now properly detected and logged in Lua Output.
- Bug Fixes: Lock and Unlock controls now working in Android and iOS Control4 apps.
- Bug Fixes: Offline password unlock, Motion Detection events are now properly triggering push notifications.

v1.0.9

- Fixed an issue with low battery notifications where MQTT was not re-enabled after VID changes.
- Minor bug fixes and improvements.

v1.0.10

- Updated Control4 Camera Programming Options
- **Device Events**: Added "System Is Locked"
- **Commands Removed**: Pan Left, Pan Right, Pan Scan, Tilt Up, Tilt Down, Tilt Scan, Zoom In, Zoom Out, Preset, Home
- **Commands Added**: Take Snapshot, Initialize Camera *, Unmute Mic, Mute Mic, Turn up speaker volume, Turn down speaker volume, Sensitivity =
- **DF511 Door Lock Commands Removed**: Toggle Door Lock
- **Conditions Added**: If Speaker volume is 1-10, If battery is <,> or =, If sensing motion, If not sensing motion, If Mic is muted, If Mic is unmuted, Sensitivity =

v1.0.11

- Updates- Added device variable to display **FaceName** in face detection notifications.
- How to Display Face Name in Push Notifications

1. Open **Control4 Composer Pro**
2. Navigate to **Agents** → **Notifications**
3. Select or create a notification for the **Face Detected** event
4. In the notification message body, click **Add Variable**
5. In the variable picker tree, expand your **DF511 Door Lock** device
6. Expand **Device Variables**

7. Select **FaceName** from the list
8. Click **OK**

- Fixes- Resolved issue where the **lock UI was taking time to load** in the updated Control4 app.

Support

Driver Version: 2 **Manufacturer:** Slomins **Model:** DF511-SL **Control4 OS:** 3.3.2+

For issues check **Lua Output logs** in Composer Pro.

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